

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 90-217

WASTE DISCHARGE REQUIREMENTS
FOR
CALIFORNIA DEPARTMENT OF CORRECTIONS
PLANNING AND CORRECTIONS DIVISION
CALIFORNIA STATE PRISON, WASCO
KERN COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The California Department of Corrections, Planning and Corrections Division, (hereafter Discharger) submitted a Report of Waste Discharge, dated 28 July 1989, for a proposed discharge from a medium-security level men's prison to be constructed near Wasco, Kern County. The treatment and disposal facility is in Section 8, T27S, R24E, MDB&M, as shown on Attachment A, which is a part of this Order.
2. The prison will be constructed as a 2,450 bed facility, but may reach design capacity of 190%, or 4,655 inmates. The prison will generate domestic wastewater with average flows of 0.81 mgd at design treatment capacity. Maximum daily flow capacity of the treatment facility is 1.0 mgd. Disposal/reclamation capacity, as determined by a hydrologic balance, is greater than 0.81 mgd.
3. The treatment scheme will include a mechanically cleaned bar screen and two, 4-million gallon capacity aerated lagoons followed by two, 2.5-million gallon capacity sedimentation ponds. The sedimentation ponds will alternately be taken off line annually for cleaning. Sludge will be spread on property controlled by the Discharger.
4. A single-lined pond will be used to store treated wastewater for land application during the crop growing season. A second unlined storage pond will be used for additional storage of wastewater as needed. The combined capacity of the storage ponds will be 380 acre feet. Typically, treated wastewater storage will increase between October and March and decrease during the remainder of the year as crop irrigation needs are met. The ponds will be protected from overtopping by an emergency spillway. The ponds will have a maximum depth of about 14 feet when full and will cover a total area of approximately 35 acres.
5. Wastewater disposal will be accomplished by flood irrigation of about 230 acres of alfalfa. Irrigation tailwater will be captured and returned to the fields.
6. Storage pond capacity is based on an annual precipitation using a return period of 25 years. Based on engineering calculations, the Discharger will retain pond overflow from heavier precipitation on-site. Although temporal ponding may occur, it will be confined to an undeveloped area of the prison and the crop growing area.

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7. The California Department of Health Services has established statewide reclamation criteria in Title 22, California Code of Regulations (CCR), Division 4, Chapter 3 (hereafter Title 22) for the use of reclaimed water.
8. The treatment and disposal facility is not located within a 100-year flood plain. Soils within the reclamation area consist predominately (98%) of the Kimberlina and Wasco series, which contain sandy loam and exhibit moderate internal drainage.
9. Depth to first ground water is approximately 200 to 250 feet below ground surface. Regional ground water generally flows to the north, but may vary seasonally depending on ground water pumping in the area. Local water quality is marginal (EC = 1435 umhos/cm, $\text{NO}_3 = 18 \text{ mg/l}$). The beneficial uses of ground water in this area are domestic and agricultural supply.
10. Surface water drainage is to Poso Creek, about five miles northeast of the proposed site. The site lies within the Semitropic Hydrologic Area (No. 558.70) in the South Valley Floor Hydrologic Unit, as defined on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
11. The Board, on 21 August 1975, adopted a Water Quality Control Plan for the Tulare Lake Basin (5D), hereafter "Basin Plan". These requirements are consistent with that Plan.
12. The California Department of Corrections has certified a final environmental impact report (EIR) in accordance with the California Environmental Quality Act (Public Resources Code Section 21000, et seq.) and the State Guidelines.
13. The Regional Board has reviewed the findings of the EIR and concurs that the project, as approved by the Department of Corrections, will not have a significant effect on water quality.
14. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
15. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

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IT IS HEREBY ORDERED that the California Department of Corrections, Planning and Corrections Division, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following at the above-described facility:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Title 23, CCR, Sections 2521(a) and 2522(a) is prohibited.

B. Discharge Specifications:

1. The monthly average daily discharge flow shall not exceed 0.81 million gallons.
2. The dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
3. The effluent from the treatment facility shall not contain constituents in excess of the following:

<u>Constituent</u>	<u>Unit</u>	<u>Mean</u>	<u>Maximum</u>
BOD ₅ ¹	mg/l	40	80
Settleable Solids	ml/l	0.2	0.5

¹ Five-day, 20° Celsius biochemical oxygen demand

4. The maximum specific electrical conductivity (EC) of the discharge shall not exceed the average EC of the source water plus 500 umhos/cm, or a maximum of 900 umhos/cm, whichever is less.
5. The discharge shall not have a pH less than 6.5 or greater than 9.0.

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6. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
7. Ponds shall be managed to prevent breeding of mosquitos. In particular,
 - a. An erosion control program shall assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
8. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
9. Irrigation or impoundment of wastewater shall not occur within 500 feet of any domestic well or within 100 feet of any irrigation well unless it is demonstrated to the satisfaction of the Executive Officer that a lesser distance is justified.
10. Use of reclaimed wastewater shall be limited to flood irrigation of fodder, fiber, and seed crops.
11. Application of wastewater to the reclamation area shall not exceed, including the nutritive value of organic and chemical fertilizers, what is reasonably necessary for the crop.
12. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
13. The storage ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 25 years, distributed monthly in accordance with historical rainfall patterns. Freeboard in all ponds shall never be less than 2 feet (measured vertically).

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C. Sludge Disposal

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Title 23, CCR, Section 2510, et seq., (Subchapter 15), and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 14 days in advance of the change.

D. Receiving Water Limitations

The Discharge, in combination with other sources, shall not cause underlying ground water to:

1. Contain waste constituents in concentrations statistically greater than the receiving water limits, where specified below, or background water quality where not specified. (For purposes of comparison, background water quality shall be determined when background monitoring provides sufficient data. Quality determined in this manner establishes "water quality protection standards.")
2. Exceed an annual average incremental increase in specific electrical conductivity greater than 6 umhos/cm, based on the most recent five-year period, or a maximum of 1600 umhos/cm.
3. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in Title 22, CCR, Division 4, Chapter 15.
4. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven day period.
5. Exceed concentrations of radionuclides specified in Title 22, CCR, Division 4, Chapter 15.
6. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.
7. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

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E. Provisions

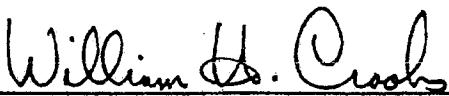
1. The Discharger shall use the best practicable cost-effective control technique currently available to limit mineralization and comply with salinity limits specified in this Order.
2. The Discharger shall submit technical reports as directed by the Executive Officer.
3. The Discharger shall comply with the attached Monitoring and Reporting Program No. 90-217, and any revisions thereto as ordered by the Executive Officer.
4. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 July 1990, which are a part of this Order.
5. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
6. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
7. If the Discharger intends to reclaim wastewater on crops other than those specified in an accepted land management plan, it shall first submit a written report demonstrating, to the satisfaction of the Executive Officer, that management of reclaimed water and irrigated properties will assure compliance with the terms of this Order.
8. In addition to modification of this Order under circumstances described in Standard Provision A.3, this Order may be modified if sludge use or disposal practices change, if applicable management practices or numerical limitations for pollutants in sewage sludge necessitate it, if there is endangerment to human health or the environment, or if the Board elects to review and revise the Order on its own motion.
9. The Discharger shall provide a letter report within 60 days of initial operation of the facility, identifying the names, titles, and certificate grades of persons supervising and operating the facility, as required by Standard Provision E.1, and identifying who to contact in an emergency.

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10. The Discharger shall submit the technical reports described in Standard Provisions A.9 and B.2 and a copy of the operation and maintenance manual for the facility within 60 days of the effective date of this Order.
11. The Discharger shall optimize reclamation of wastewater and minimize the volume of wastewater discharged to the unlined pond. The Discharger shall notify this office prior to using the unlined storage pond to store wastewater.
12. The Discharger must comply with all conditions of this Order. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in modification or revocation of this Order.
13. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 10 August 1990.


WILLIAM H. CROOKS, Executive Officer

JGW:cjs:8/10/90

Attachments

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 90-217

FOR
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Specific sample station locations shall be established under direction of the Board's staff and a description of the stations shall be attached to this Order.

INFLUENT MONITORING

Samples shall be collected at approximately the same time as effluent samples and should be representative of the influent for the period sampled. The following shall constitute the influent monitoring program.

<u>Constituent</u>	<u>Unit</u>	<u>Type of Monitoring/ Measurement</u>	<u>Monitoring Frequency</u>
Flow	MGD	Estimate	Daily
Settleable Solids	ml/l	Grab	Weekly
Total Suspended Solids	mg/l	Grab	Monthly
BOD ₅	mg/l	Grab	Monthly

EFFLUENT MONITORING

Effluent samples (except for Dissolved Oxygen) shall be collected just prior to discharge to the storage ponds. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from storage ponds will be considered adequately composited. Time of collection of a grab sample shall be recorded. The following shall constitute the effluent monitoring program:

<u>Constituent</u>	<u>Unit</u>	<u>Type of Monitoring/ Measurement</u>	<u>Monitoring Frequency</u>
Dissolved Oxygen ²	mg/l	Grab	Weekly
pH	pH units	Grab	Weekly
Settleable Solids	ml/l	Grab	Weekly
BOD ₅	mg/l	8-hr. Composite	2/Month
Total Suspended Solids	mg/l	8-hr. Composite	Monthly

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<u>Constituent</u>	<u>Unit</u>	<u>Type of Monitoring/ Measurement</u>	<u>Monitoring Frequency</u>
Kjeldahl Nitrogen (as N)	mg/l	Grab	Monthly
Nitrate (as N)	mg/l	Grab	Monthly
Specific Electrical Conductivity @ 20° C	umhos/cm	Grab	Quarterly

- ¹ If results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly mean for BOD), or indicate a violation and potential upset of the treatment process, the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.
- ² Samples shall be collected from each pond opposite of the inlet side and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

POND MONITORING

The freeboard shall be monitored on all storage ponds in use to the nearest half-foot. The following shall constitute the freeboard monitoring program:

<u>Constituent</u>	<u>Unit</u>	<u>Type of Sample</u>	<u>Frequency</u>
Freeboard	feet	Observation	Weekly

Permanent markers shall be placed in each pond with calibrations indicating the water level at design capacity and available operational freeboard.

In addition, the Discharger shall inspect the condition of the ponds once per week and write visual observations in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, grey, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary,

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the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

SOURCE WATER MONITORING

A representative sample shall be collected from the source which supplies water to the facility according to the specified frequency.

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
EC ¹	umhos/cm	Grab	Quarterly

- ¹ If the source water is from more than one source, the EC shall be reported as the weighted average and include copies of supporting calculations.

SLUDGE MONITORING

When sludge is removed from ponds, but prior to disposal, a composite sample shall be analyzed, on a dry weight basis, for Total Solids (%), Nitrogen (total, $\text{NH}_4\text{-N}$, and $\text{NO}_3\text{-N}$), Total Phosphorous, Total Potassium, Total PCBs, and totals of specific metals (Pb, Zn, Cu, Ni, Cd, and Ag). Analytical results shall be submitted to the Executive Officer. Analysis of soluble concentrations of these specific metals shall also be included. If final disposal is proposed to go to land, a technical report analyzing application rates and procedures relative to Department of Health Services' *Manual of Good Practices for Landspreading of Sewage Sludge* and EPA's *Process Design Manual for Land Application of Municipal Sludges* and Title 23, Section 2511(f) of the California Code of Regulations, shall be completed and submitted to the Executive Officer for approval.

ANNUAL LAND MANAGEMENT REPORTS

By 30 January of each year, the Discharger shall submit an annual summary report on the land management operation after the conclusion of the crop season. The report shall discuss total water application over the season; the total wastewater applied; the total nutrient loading from wastewater, sludges, and chemical fertilizers; and amount of nutrients removed through harvest of crop. In short, the report shall present a mass balance relative to pollutants of concern and hydraulic loading.

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REPORTING

Monthly monitoring reports shall be submitted to the Regional Board by the 15th day of the following month.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in a manner that illustrates clearly whether the discharge complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By 31 January of each year, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names and general responsibilities of persons operating and maintaining the wastewater treatment plant.
- b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.
- c. A statement identifying whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
- d. A statement certifying when the flow meter and other monitoring devices were last calibrated, including identification of who performed the calibration (Standard Provision C.4.).
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

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All reports submitted in response to this Order shall comply with the signatory requirements of Standard Provision B.3.

The Discharger shall implement this monitoring program beginning the first day of the month following the date of signature, below, or upon initiation of discharge, whichever is later.

Ordered by William H. Crooks
WILLIAM H. CROOKS, Executive Officer

10 August 1990
(Date)

JGW:cjs:8/10/90

INFORMATION SHEET

CALIFORNIA DEPARTMENT OF CORRECTIONS
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The California Department of Corrections filed a Report of Waste Discharge for the discharge of domestic wastewater from a proposed 2,450 bed medium-security prison for men in Kern County. Inmate population may reach as high as 4,655. The prison will employ approximately 880 full time staff over three, eight-hour shifts. The site is three miles west of Wasco and immediately south of State Highway 46 in Section 8, T27S, R24E, MDB&M.

The prison will generate domestic wastewater with average flows of 0.81 mgd at design treatment capacity. Maximum daily flow capacity of the treatment facility is 1.0 mgd. Disposal/reclamation capacity, as determined by a hydrologic balance, is greater than 0.81 mgd.

The treatment scheme will include a mechanically cleaned bar screen and two, 4- million gallon capacity aerated lagoons followed by two, 2.5-million gallon capacity sedimentation ponds. The sedimentation ponds will alternately be taken off line annually for cleaning. Sludge will be spread on property controlled by the Discharger.

A single-lined pond will be used to store treated wastewater for land application during the crop growing season. A second unlined storage pond will be used for additional storage of wastewater as needed. The combined capacity of the storage ponds will be 380 acre feet. Typically, treated wastewater storage will increase between October and March and decrease during the remainder of the year as crop irrigation needs are met. The ponds will be protected from overtopping by an emergency spillway. The ponds will have a maximum depth of about 14 feet when full and will cover a total area of approximately 35 acres.

Storage pond capacity is based on an annual precipitation using a return period of 25 years. Based on engineering calculations, the Discharger will retain pond overflow from the heavier precipitation onsite. Although partial temporal ponding may occur, it will be confined to an undeveloped area of the prison and the crop growing area.

Wastewater disposal will be accomplished by flood irrigation of about 230 acres of alfalfa. Irrigation tail water will be captured and returned to the fields.

The treatment and disposal facility is not within a 100-year flood plain. Soils within the reclamation area consist predominately (98%) of the Kimberlina and Wasco series, which contain sandy loam and exhibit moderate internal drainage.

Depth to first ground water is approximately 200 to 250 feet below ground surface. Regional ground water generally flows to the north, but may vary

INFORMATION SHEET - Continued

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seasonally depending on ground water pumping in the area. Local water quality is marginal (EC = 1435 umhos/cm, NO_3 = 18 mg/l). The beneficial uses of ground water in this area are domestic and agricultural supply. Surface water drainage is to Poso Creek, about five miles northeast of the proposed site.

Ground water monitoring is not required by the proposed Order because of the level of treatment, reclamation, and hydrogeologic characteristics of the area. If evidence indicates later that ground water monitoring may be appropriate, or the Discharger relies heavily on disposal of wastewater in the unlined pond, ground water monitoring will be added then by the Executive Officer.

The California Department of Corrections has certified a final environmental impact report (EIR) in accordance with the California Environmental Quality Act (Public Resources Code Section 21000, et seq.) and the State Guidelines. The Regional Board staff reviewed the findings of the EIR and concurs that the project, as approved by the Department of Corrections, will not have a significant effect on water quality.

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